

Module 1

Computing in GCP



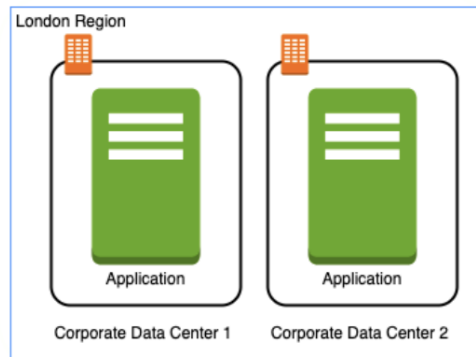
Goals: Understand about virtual machines, machine images, persistent disk.

- Create instance
- Regions and zones
- Type of instance
- Additional disks
- Disk snapshots
- Attach and detach additional disks

- Imagine that your application is deployed in a data centre in London What would be the challenges?
- Challenge 1 : Slow access for users from other parts of the world (high latency)
- Challenge 2 : What if the data centre crashes? Your application goes down (low availability)

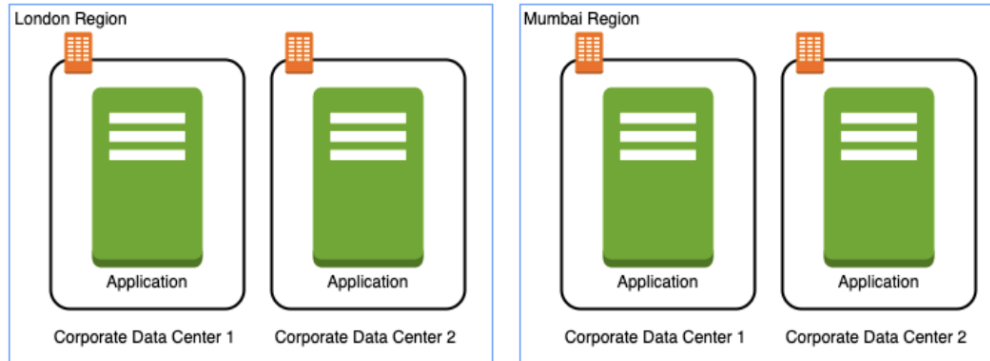


Multiple data centers



- Let's **add in one more data center** in London
- What would be the challenges?
 - Challenge 1 : Slow access for users from other parts of the world
 - Challenge 2 (**SOLVED**) : What if one data center crashes?
 - Your application is **still available** from the other data center
 - Challenge 3 : What if **entire region** of London is unavailable?
 - Your application goes down

- Let's add a new region : Mumbai What would be the challenges? Challenge 1 (PARTLY SOLVED) : Slow access for users from other parts of the world You can solve this by adding deployments for your applications in other regions
- Challenge 2 (SOLVED) : What if one data center crashes? Your application is still live from the other data centers
- Challenge 3 (SOLVED) : What if entire region of London is unavailable? Your application is served from Mumbai



Regions

Imagine setting up data centers in different regions around the world
Would that be easy ?

(Solution) Google provides 20+ regions around the world Expanding every year
Region : Specific geographical location to host your resources

Advantages: High Availability Low Latency Global Footprint Adhere to government regulations



Zones

1. How to achieve high availability in the same region (or geographic location)?
 - Enter **Zones**
2. Each Region has three or more **zones**
 - (Advantage) **Increased availability and fault tolerance** within same region



Virtual machine

- ▶ Google Compute Engine is Google's Infrastructure-as-service (IaaS) virtual machine offering.
- ▶ Customers can configure and run a wide variety of virtual machine configurations.
- ▶ VM Instances are comprised of an Operating System and infrastructure resources such as CPU, Memory, Disk, and Networking

Machine types

- ▶ Standard machine type

Ideal for typical balanced instances with respect to RAM and CPU

- ▶ High-memory machine types

Ideal for applications that require more memory

- ▶ Large machine types

Ideal for resource-intensive workloads

- ▶ Custom machine types

Machine name	vCPUs ¹	Memory (GB)	Max number of persistent disks (PDs) ²	Max total PD size (TB)	Local SSD	Maximum egress bandwidth (Gbps) ³
e2-standard-2	2	8	128	257	No	4
e2-standard-4	4	16	128	257	No	8
e2-standard-8	8	32	128	257	No	16
e2-standard-16	16	64	128	257	No	16
e2-standard-32	32	128	128	257	No	16

Images

- Images contain a bootloader, Operating System, file system structure, and any software customizations needed for your deployment.
- Google Compute Engine (GCE) uses the selected image to create a persistent boot disk for each instance.

[PUBLIC IMAGES](#) [CUSTOM IMAGES](#) [SNAPSHOTS](#) [ARCHIVE SNAPSHOTS](#)

Operating system

Debian

Version *

Debian GNU/Linux 11 (bullseye)

x86/64, amd64 built on 20230711

Boot disk type *

Balanced persistent disk

[COMPARE DISK TYPES](#)

Size (GB) *

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Disks

- All Compute Engine needs Disk to read/write data or to operate.
- Persistent Disk - Single root disk attached to VM.
 - Common and Default Option
 - Not Directly attached to Compute Engine, but Network Attached Disk
 - Standard and SSD options available
- Local SSD - Directly attached to Compute Engine VM

➤ Persistent Disks

- Very flexible and Powerful.
- Independent from VM not physically attached.
- Can be Detach/Move.
- Can be read multiple instance at Once.
- Preserve Data even after deleting Instance.
- Resize, Move, Attach additional Disk- even when in-use.

```
sudo resize2fs /dev/sda1
```

- SSD Options are available.
- Encryption - Google Encryption or Custom Encryption.

➤ Local SSD

- Highest Performance Disk
- Physically attached to VM
- Can't be boot disk
- Must create on Instance Creation
- All Data lost in case Instance Terminate
- Only Support Google Supplied Encryption.
- Can attach Local SSD and Persistent Disk to same Instance

Persistent disk snapshot

- A GCP persistent disk snapshot is a point-in-time copy of a persistent disk. Snapshots can be used to back up data, restore data, or create new disks.
- There are three types of GCP persistent disk snapshots:
 - ✓ Standard snapshots: These snapshots are stored in the default storage location for your project. They are the most common type of snapshot.
 - ✓ Instant snapshots: These snapshots are created very quickly, but they are only stored in memory for a short period of time. They are useful for creating temporary backups or for restoring data from a failed VM instance.
 - ✓ Archive snapshots: These snapshots are stored in a low-cost storage location. They are a good option for long-term backups.
- To create a GCP persistent disk snapshot, you can use the gcloud command-line tool or the Google Cloud Console. Here is an example of how to create a standard snapshot using the gcloud command-line tool:
`gcloud compute snapshots create my-snapshot --source-disk my-disk`

Preemptible VM

- **Short-lived cheaper** (up to 80%) compute instances
- Can be stopped by GCP any time within 24 hours
- Instances get 30 second warning (to save anything they want to save) **Use case:**
- Your applications are **fault tolerant**
- You are very **cost sensitive**
- Your workload is **NOT immediate**
- Example: Non immediate batch processing jobs
- **RESTRICTIONS:**
- NOT always available
- NO SLA and CANNOT be migrated to regular VMs
- NO Automatic Restarts

Quizzes

- What are the pre-requisites to be able to create a VM instance?
- How to access to your VM instance ?
- You do not want to expose a VM to internet

Thank you

